

Q6) Write a C program for ciphertext has been generated with an affine cipher. The most frequent letter of the ciphertext is “B,” and the second most frequent letter of the ciphertext is “U”. Break this code.

Aim: To write a C program to break an **Affine Cipher** using **frequency analysis**, assuming that the most frequent ciphertext letter corresponds to 'E' and the second most frequent corresponds to 'T', the two most common letters in English.

Algorithm:

1. Start the program.
2. Read the ciphertext from the user.
3. Count the frequency of each alphabet character in the ciphertext.
4. Identify the most frequent and second most frequent ciphertext letters.
5. Assume that the most frequent letter corresponds to 'E' and the second most frequently corresponds to 'T'.
6. Try all valid values of **a** (coprime with 26) and all values of **b** (0–25).
7. Decrypt the ciphertext using each possible key pair (**a, b**).
8. Display all possible plaintexts and allow the user to identify the meaningful message.
9. Stop the program.

Result:

Thus, the C program to break an **Affine Cipher** using **frequency analysis** was successfully executed. By assuming that **B** and **U** correspond to the common English letters **E** and **T**, the probable encryption key was determined as **a = 3** and **b = 15**, enabling successful decryption of the ciphertext.

Code:

```
1 #include <stdio.h>
2 #include <string.h>
3 #include <ctype.h>
4 int gcd(int a, int b)
5 {
6     while(b != 0)
7     {
8         int t = b;
9         b = a % b;
10        a = t;
11    }
12    return a;
13 }
14 int modInverse(int a)
15 {
16     int i;
17     for(i = 1; i < 26; i++)
18     {
19         if((a * i) % 26 == 1)
20             return i;
21     }
22     return -1;
23 }
24 int main()
25 {
26     char cipher[100], plain[100];
27     int a, b, i, inv;
28     printf("Enter Ciphertext: ");
29     scanf("%s", cipher);
30     printf("\nPossible Plaintexts:\n\n");
```

Line: 19 Col: 30 Sel: 0 Lines: 56 Length: 1223 Insert Done parsing in 0.015 seconds

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Output:

```
C:\Users\SAIL\Desktop\Lab4.exe
Enter Ciphertext: Buub
Possible Plaintexts:
a= 1 b= 0 --> BUUB
a= 1 b= 1 --> ATTA
a= 1 b= 2 --> ZSSZ
a= 1 b= 3 --> YRRY
a= 1 b= 4 --> XQQX
a= 1 b= 5 --> WPPW
a= 1 b= 6 --> VOOV
a= 1 b= 7 --> UNNU
a= 1 b= 8 --> TMMT
a= 1 b= 9 --> SLLS
a= 1 b=10 --> RKKR
a= 1 b=11 --> QJJQ
a= 1 b=12 --> PIIP
a= 1 b=13 --> OHHO
a= 1 b=14 --> NGGN
a= 1 b=15 --> MFFM
a= 1 b=16 --> LEEL
a= 1 b=17 --> KDDK
a= 1 b=18 --> JCCJ
a= 1 b=19 --> IBBI
a= 1 b=20 --> HAAH
a= 1 b=21 --> GZZG
a= 1 b=22 --> FYYF
a= 1 b=23 --> EXXE
a= 1 b=24 --> DWWD
a= 1 b=25 --> CVVC
```